CLOSED-FORM GAUSSIAN INFORMATION PROJECTION FOR EMBEDDED HYPERCUBES AND BETA–GAUSSIAN MIXTURES

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Abstract. We derive closed-form Gaussian information projections of Beta–Gaussian distributions supported on embedded hypercubes, with the optimal Gaussian characterized by moment matching. Extending this construction to finite embedded cubical complexes, we establish a mixture approximation bound that is independent of the embedding geometry and ambient dimension. In the uniform case, the bound reduces to a linear term in the intrinsic dimension.

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1. Introduction

Gaussian distributions provide a tractable representation of probability measures, but their approximation quality can deteriorate when the underlying distribution is supported on a bounded or geometrically constrained domain. This motivates the study of Gaussian information projections for distributions defined on embedded geometric structures.

In this work, we consider affine embeddings of the unit hypercube into an ambient Euclidean space. We equip the reference hypercube with product-Beta distributions and introduce Gaussian perturbations in the orthogonal complement of the embedding's tangent space, obtaining absolutely continuous probability distributions in the ambient space. We study their forward Kullback–Leibler information projections onto the family of nondegenerate Gaussian distributions.

Our main result provides a closed-form characterization of the unique optimal Gaussian through moment matching. Moreover, the resulting minimum divergence depends only on the intrinsic Beta parameters, independently of the embedding geometry and ambient dimension. For the uniform case, the divergence reduces to a linear function of the intrinsic dimension.

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We further extend this construction to finite, pure embedded cubical complexes. By combining componentwise Gaussian projections with the joint convexity of the Kullback–Leibler divergence, we establish an explicit upper bound for the resulting Gaussian-mixture approximation. In the uniform case, this bound is again linear in the intrinsic dimension and independent of the geometry and ambient dimension.

2. Construction

Definition 2.1 (Embedded Hypercube)**.** *For $1 \le d \le D$, an embedded d-cube in $\mathbb{R}^D$ is*

$$C = v + E[0,1]^d, \qquad v \in \mathbb{R}^D, \quad E \in \mathbb{R}^{D\times d}, \quad \operatorname{rank}(E) = d. \tag{2.1}$$

Its tangent and normal spaces are

$$T_C = \operatorname{range}(E), \qquad N_C = T_C^{\perp}. \tag{2.2}$$

Remark 2.2. *An embedded hypercube is understood affinely; its image is a d-dimensional parallelotope, not necessarily an orthogonal cube.*

Proposition 2.3. *Let $C = F_C([0,1]^d)$, where $F_C(x) = v + Dx$ and $D \in \mathbb{R}^{n\times d}$ has full column rank. Then*

$$\operatorname{vol}_d(C) = \sqrt{\det(D^\top D)}. \tag{2.3}$$

Proof. By the area formula (see [1, Theorem 3.2.3]),

$$\operatorname{vol}_d(C) = \int_{[0,1]^d} \sqrt{\det(D^\top D)}\, dx = \sqrt{\det(D^\top D)}. \tag{2.4}$$

□

Definition 2.4 (Faces of an Affine Cube)**.** *Let $C = v + E[0,1]^d$ be an embedded affine d-cube. For $I \subseteq \{1,\dots,d\}$ and $\epsilon \in \{0,1\}^I$, define*

$$F_{I,\epsilon}(C) \coloneqq v + E\big\{x \in [0,1]^d : x_i = \epsilon_i \text{ for all } i \in I\big\}. \tag{2.5}$$

These sets are the faces of C, with

$$\dim F_{I,\epsilon}(C) = d - |I|. \tag{2.6}$$

Definition 2.5 (Embedded Cubical Complex)**.** *A finite collection $\mathcal{K}$ of embedded affine cubes is an* embedded cubical complex *if*

(1) $F \in \mathcal{K}$ whenever F is a face of $C \in \mathcal{K}$;
(2) $C \cap C'$ is either empty or a face of both $C, C' \in \mathcal{K}$.

The complex is pure of dimension d if

$$\dim C = d \quad \text{for every inclusion-maximal } C \in \mathcal{K}. \tag{2.7}$$

Its realization and maximal cells are denoted by

$$|\mathcal{K}| \coloneqq \bigcup_{C\in\mathcal{K}} C, \quad \mathcal{K}^{[d]} \coloneqq \{C \in \mathcal{K} : \ C \text{ is inclusion-maximal}\}. \tag{2.8}$$

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Proposition 2.6 (Cubical Volume Decomposition)**.** *Let $\mathcal{K}$ be a finite, pure, d-dimensional embedded cubical complex in $\mathbb{R}^D$. Then*

$$\mathcal{H}^d(|\mathcal{K}|) = \sum_{C \in \mathcal{K}^{[d]}} \mathrm{vol}_d(C). \tag{2.9}$$

Moreover, if $0 < \mathcal{H}^d(|\mathcal{K}|) < \infty$, then

$$\mathcal{U}_{|\mathcal{K}|} = \sum_{C \in \mathcal{K}^{[d]}} \frac{\mathrm{vol}_d(C)}{\mathcal{H}^d(|\mathcal{K}|)} \mathcal{U}_C, \qquad \mathcal{U}_A := \frac{\mathcal{H}^d|_A}{\mathrm{vol}_d(A)}. \tag{2.10}$$

Proof. Distinct maximal cells intersect in proper faces, hence in $\mathcal{H}^d$-null sets. The result follows from finite additivity of $\mathcal{H}^d$. □

Definition 2.7 (Product-Beta density)**.** *For parameter vectors $a = (a_1, \dots, a_d) \in (0, \infty)^d$ and $b = (b_1, \dots, b_d) \in (0, \infty)^d$, let $X = (X_1, \dots, X_d)^\top$ have independent coordinates*

$$X_i \sim \mathrm{Beta}(a_i, b_i), \qquad i = 1, \dots, d. \tag{2.11}$$

Its density with respect to Lebesgue measure on $[0,1]^d$ is

$$f_{a,b}(x) = \prod_{i=1}^{d} \frac{x_i^{a_i - 1}(1 - x_i)^{b_i - 1}}{B(a_i, b_i)}, \qquad x \in (0,1)^d, \tag{2.12}$$

where $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ is the beta function. The density is set to zero outside $[0,1]^d$.

Proposition 2.8 (Moments and entropy of a product-Beta law)**.** *Under definition 2.7, define, for each i,*

$$\mu_i = \frac{a_i}{a_i + b_i}, \qquad s_i^2 = \frac{a_i b_i}{(a_i + b_i)^2 (a_i + b_i + 1)}. \tag{2.13}$$

Then

$$\mathbb{E}[X] = \mu, \qquad \mathrm{Cov}(X) = V := \mathrm{diag}(s_1^2, \dots, s_d^2) \succ 0. \tag{2.14}$$

The differential entropy is finite and equals

$$h(X) = \sum_{i=1}^{d} h_i, \tag{2.15}$$

where

$$\begin{aligned} h_i = {} & \log B(a_i, b_i) - (a_i - 1)\psi(a_i) - (b_i - 1)\psi(b_i) \\ & + (a_i + b_i - 2)\psi(a_i + b_i), \end{aligned} \tag{2.16}$$

and ψ is the digamma function.

Proof. The moment identities are the standard univariate Beta moment formulas, together with independence. For entropy, taking the expectation of the logarithm of the density gives

$$\begin{aligned} -h(X_i) &= \mathbb{E}[\log f_{X_i}(X_i)] \\ &= -\log B(a_i, b_i) + (a_i - 1)\mathbb{E}[\log X_i] + (b_i - 1)\mathbb{E}[\log(1 - X_i)]. \end{aligned}$$

Differentiating $\log B(a,b)$ with respect to its two arguments, or equivalently differentiating the beta integral under the integral sign, gives

$$\mathbb{E}[\log X_i] = \psi(a_i) - \psi(a_i + b_i), \quad \mathbb{E}[\log(1 - X_i)] = \psi(b_i) - \psi(a_i + b_i).$$

Substitution yields eq. (2.16). These expectations are finite for all strictly positive parameters: near either endpoint, $x^{a_i-1}|\log x|$ and $(1-x)^{b_i-1}|\log(1-x)|$ are integrable. Thus each marginal entropy is finite. Independence gives additivity of entropy and eq. (2.15). □

Definition 2.9 (Beta law on an embedded hypercube)**.** *Let $C = v + E[0,1]^d$ be an embedded hypercube. For $a, b \in (0,\infty)^d$, the* embedded product-Beta law *on C is the pushforward of the product-Beta law in definition 2.7 under $F_C(x) = v + Ex$. In other words,*

$$Y_T = v + EX, \qquad X_i \overset{\text{ind}}{\sim} \text{Beta}(a_i, b_i). \tag{2.17}$$

We denote its law by $P_T(C; a, b)$. Its intrinsic density is understood with respect to $\mathcal{H}^d$ on the affine hull of C.

Proposition 2.10 (Embedded moments and intrinsic entropy)**.** *Under definition 2.9,*

$$m_T := \mathbb{E}[Y_T] = v + E\mu, \tag{2.18}$$

$$\Sigma_T := \text{Cov}(Y_T) = EVE^\top. \tag{2.19}$$

The covariance is positive definite on T_C and zero on N_C. The intrinsic differential entropy is

$$h_{T_C}(Y_T) = \sum_{i=1}^{d} h_i + \frac{1}{2}\log\det(E^\top E). \tag{2.20}$$

Proof. The moment formulas follow from the affine transformation rule. Since E has full column rank and $V \succ 0$, the restriction of $EVE^\top$ to $T_C = \text{range}(E)$ is positive definite. For the entropy, the constant d-dimensional Jacobian of F_C is $J_C = \sqrt{\det(E^\top E)}$. The change-of-variables formula on the affine hull gives $h_{T_C}(Y_T) = h(X) + \log J_C$, which is eq. (2.20). □

Definition 2.11 (Beta–Gaussian distribution)**.** *Let $C = v + E[0,1]^d \subseteq \mathbb{R}^D$ be an embedded hypercube, with $d \le D$. Choose matrices $U_T \in \mathbb{R}^{D\times d}$ and $U_N \in \mathbb{R}^{D\times(D-d)}$ whose columns are orthonormal bases of T_C and N_C, respectively. Let*

$$X_i \overset{\text{ind}}{\sim} \text{Beta}(a_i, b_i), \qquad Z_N \sim \mathcal{N}_{D-d}(\nu_N, S_N), \qquad S_N \succ 0, \tag{2.21}$$

with X and Z_N independent. Define

$$W := v + EX + U_N Z_N. \tag{2.22}$$

The law of W is called a Beta–Gaussian distribution *associated with C, and is denoted by $P(C; a, b, \nu_N, S_N)$. If $d = D$, the normal term is absent and all statements involving U_N, Z_N, S_N are interpreted on the zero-dimensional space.*

Lemma 2.12 (Absolute continuity and finite entropy)**.** *The Beta–Gaussian random vector in definition 2.11 is absolutely continuous with respect to D-dimensional Lebesgue measure and has finite differential entropy. More precisely, for $d < D$,*

$$h(W) = \sum_{i=1}^{d} h_i + \frac{1}{2}\log\det(E^\top E) + \frac{D-d}{2}\log(2\pi e) + \frac{1}{2}\log\det S_N. \tag{2.23}$$

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Proof. The product-Beta vector has a density on $(0,1)^d$ and finite entropy by proposition 2.8. Its affine image has finite intrinsic entropy by proposition 2.10. The Gaussian normal coordinate has a nondegenerate density and finite entropy. Under the orthogonal affine coordinate map

$$\Phi(w) = \begin{pmatrix} U_T^\top (w - v) \\ U_N^\top (w - v) \end{pmatrix},$$

the absolute Jacobian determinant is one and

$$\Phi(W) = \begin{pmatrix} U_T^\top EX \\ Z_N \end{pmatrix}.$$

The two blocks are independent. Their product law is absolutely continuous in $\mathbb{R}^D$, and entropy additivity gives eq. (2.23). The same argument without the normal block applies when $d = D$. □

Remark 2.13. *The normal Gaussian is essential when $d < D$: without it the distribution is supported on a d-dimensional affine subspace and is singular with respect to ambient Lebesgue measure. The positive-definite normal covariance makes the ambient density nondegenerate. No isotropy assumption is imposed on S_N.*

Definition 2.14 (Forward Kullback–Leibler divergence)**.** *For Lebesgue densities p, q on $\mathbb{R}^D$, define*

$$\mathrm{KL}(P : Q) = \int_{\mathbb{R}^D} p(x) \log \frac{p(x)}{q(x)}\, dx, \tag{2.24}$$

with the usual extended-real convention. Let $\mathcal{G}_D$ denote the family of nondegenerate Gaussian densities on $\mathbb{R}^D$. The information projection of P onto $\mathcal{G}_D$ is any minimizer of $\mathrm{KL}(P : Q)$ over $Q \in \mathcal{G}_D$.

Proposition 2.15 (Gaussian information projection by moment matching)**.** *Let P be absolutely continuous on $\mathbb{R}^D$, have finite differential entropy, finite second moments, and positive-definite covariance Σ_P. Then its unique information projection onto $\mathcal{G}_D$ is*

$$Q^* = \mathcal{N}_D(\cdot \mid m_P, \Sigma_P), \qquad m_P = \mathbb{E}_P[X]. \tag{2.25}$$

Proof. For $Q = \mathcal{N}_D(\cdot \mid m, \Sigma)$, direct integration gives

$$\begin{aligned} \mathrm{KL}(P : Q) = & -h(P) + \frac{D}{2} \log(2\pi) + \frac{1}{2} \log\det \Sigma \\ & + \frac{1}{2} \operatorname{tr}(\Sigma^{-1}\Sigma_P) + \frac{1}{2}(m - m_P)^\top \Sigma^{-1}(m - m_P). \end{aligned} \tag{2.26}$$

For fixed $\Sigma \succ 0$, the last term is uniquely minimized at $m = m_P$. To minimize over Σ, set $A = \Sigma_P^{-1/2} \Sigma \Sigma_P^{-1/2}$. Up to constants independent of A, the objective is $\log\det A + \operatorname{tr}(A^{-1})$. If $\lambda_1, \ldots, \lambda_D$ are the positive eigenvalues of A, this is $\sum_k (\log \lambda_k + \lambda_k^{-1})$, whose unique minimum occurs at $\lambda_k = 1$ for every k (the scalar function has derivative $(\lambda - 1)/\lambda^2$ and a unique global minimum at one). Thus $\Sigma = \Sigma_P$, proving existence and uniqueness. □

Theorem 2.16 (Closed-form projection of a Beta–Gaussian law)**.** *Let $P = P(C; a, b, \nu_N, S_N)$ be the Beta–Gaussian law in definition 2.11. Define μ, V by eq. (2.13), and let $\overline{\Sigma}_N =$*

$U_N S_N U_N^\top$ (zero if $d = D$). Then the unique Gaussian information projection of P onto $\mathcal{G}_D$ is

$$Q^* = \mathcal{N}_D(\cdot \mid m, \Sigma), \tag{2.27}$$

where

$$m = v + E\mu + U_N \nu_N, \tag{2.28}$$

$$\Sigma = EVE^\top + \overline{\Sigma}_N. \tag{2.29}$$

Its minimum divergence is

$$\mathrm{KL}(P : Q^*) = C_d(a, b) := \sum_{i=1}^{d} \left[\frac{1}{2} \log(2\pi e s_i^2) - h_i \right], \tag{2.30}$$

where s_i^2 and h_i are given by eqs. (2.13) *and* (2.16)*. In particular, the minimum divergence is independent of the ambient dimension, the embedding v, E, the hypercube volume, the normal mean, and the normal covariance.*

Proof. Independence of X and Z_N gives the mean and covariance in eqs. (2.28) and (2.29). The tangential covariance is positive definite on T_C, while $\overline{\Sigma}_N$ is positive definite on N_C; since these spaces are orthogonal complements, $\Sigma \succ 0$. The entropy is finite by lemma 2.12, and the second moments are finite because the Beta component is bounded and the normal component is Gaussian. Hence proposition 2.15 applies and proves the stated unique projection.

It remains to compute the divergence. In orthonormal coordinates adapted to $T_C \oplus N_C$, both P and Q^* factor into a tangential and a normal distribution. Their normal factors agree exactly, so invariance and additivity of KL divergence give

$$\mathrm{KL}(P : Q^*) = \mathrm{KL}(P_T : Q_T^*). \tag{2.31}$$

The entropy of the tangential target is, by eq. (2.20),

$$h_{T_C}(Y_T) = \sum_{i=1}^{d} h_i + \frac{1}{2} \log\det(E^\top E). \tag{2.32}$$

Let U_T be an orthonormal basis matrix for T_C and put $R = U_T^\top E$. Since $E = U_T R$ and R is invertible, the matrix of the tangential covariance in this orthonormal basis is $RVR^\top$. Therefore

$$\begin{aligned} \det_{T_C}(\Sigma_T) &= \det(R)^2 \det(V) \\ &= \det(E^\top E) \prod_{i=1}^{d} s_i^2. \end{aligned} \tag{2.33}$$

The entropy of the moment-matched tangential Gaussian is consequently

$$h(Q_T^*) = \frac{d}{2} \log(2\pi e) + \frac{1}{2} \log\det(E^\top E) + \frac{1}{2} \sum_{i=1}^{d} \log s_i^2. \tag{2.34}$$

Subtracting the target entropy and using eq. (2.31) gives eq. (2.30). All geometric and normal terms cancel, as claimed. □

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Corollary 2.17 (Uniform distribution on an embedded hypercube)**.** *If $a_i = b_i = 1$ for all i, then X is uniform on $[0,1]^d$, $\mu_i = 1/2$, $s_i^2 = 1/12$, and $h_i = 0$. Hence*

$$m = v + \frac{1}{2}E\mathbf{1}_d + U_N\nu_N, \qquad \Sigma = \frac{1}{12}EE^\top + \overline{\Sigma}_N, \tag{2.35}$$

and

$$\mathrm{KL}(P : Q^*) = \frac{d}{2}\log\left(\frac{\pi e}{6}\right). \tag{2.36}$$

The divergence depends only on the intrinsic dimension d.

Proof. For the uniform Beta$(1,1)$ law, the stated moments and entropy follow immediately from the univariate formulas. Substitution into theorem 2.16 yields the result. □

Remark 2.18 (Nonnegativity)**.** *Each summand in eq.* (2.30) *is the KL divergence between a univariate Beta(a_i, b_i) density and its moment-matched Gaussian on $\mathbb{R}$. It is therefore nonnegative. Equality cannot occur for a nondegenerate Beta law, since its support is bounded whereas a nondegenerate Gaussian has unbounded support. Thus $C_d(a,b) > 0$ for $d \geq 1$.*

Definition 2.19 (Finite Gaussian mixture)**.** *A finite Gaussian mixture on $\mathbb{R}^D$ is a density*

$$q(x) = \sum_{j=1}^{M} \lambda_j \mathcal{N}_D(x \mid m_j, \Sigma_j), \qquad \lambda_j \geq 0, \quad \sum_{j=1}^{M} \lambda_j = 1, \quad \Sigma_j \succ 0. \tag{2.37}$$

Proposition 2.20 (Joint convexity of KL divergence)**.** *Let p_j, q_j be probability densities with respect to a common measure and let $\lambda_j \geq 0$ with $\sum_j \lambda_j = 1$. Then*

$$\mathrm{KL}\left(\sum_{j=1}^{M} \lambda_j p_j : \sum_{j=1}^{M} \lambda_j q_j\right) \leq \sum_{j=1}^{M} \lambda_j \,\mathrm{KL}(p_j : q_j), \tag{2.38}$$

where the inequality is understood in the extended-real sense.

Proof. For almost every x, apply the log-sum inequality

$$\left(\sum_j r_j\right) \log \frac{\sum_j r_j}{\sum_j s_j} \leq \sum_j r_j \log \frac{r_j}{s_j}$$

with $r_j = \lambda_j p_j(x)$ and $s_j = \lambda_j q_j(x)$, using the standard zero conventions. Integration gives eq. (2.38). □

Theorem 2.21 (Beta–Gaussian mixture approximation on an embedded cubical complex)**.** *Let $\mathcal{K}$ be a finite, pure, d-dimensional embedded cubical complex in $\mathbb{R}^D$, with maximal cells $C_1, \ldots, C_M$. For each j, write*

$$C_j = v_j + E_j[0,1]^d,$$

where E_j has full column rank. Choose parameters $a_j, b_j \in (0,\infty)^d$, a normal mean $\nu_{N,j}$, and a positive-definite normal covariance $S_{N,j}$ on N_{C_j} (with the normal term omitted if $d = D$). Let P_j be the corresponding Beta–Gaussian law and let q_j be its unique Gaussian information projection as specified by theorem 2.16. For arbitrary mixture weights

$$\lambda_j > 0, \qquad \sum_{j=1}^{M} \lambda_j = 1,$$

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define

$$P = \sum_{j=1}^{M} \lambda_j P_j, \qquad Q = \sum_{j=1}^{M} \lambda_j q_j. \tag{2.39}$$

Then each q_j is the unique Gaussian information projection of P_j, and

$$\mathrm{KL}(P : Q) \le \sum_{j=1}^{M} \lambda_j C_d(a_j, b_j), \tag{2.40}$$

where C_d is given in eq. (2.30). In particular, the bound is independent of the ambient dimension, the embeddings of the cells, and all normal means and covariance matrices.

Proof. The componentwise information-projection statement and the identity $\mathrm{KL}(P_j : q_j) = C_d(a_j, b_j)$ follow from theorem 2.16. Applying joint convexity, proposition 2.20, to the mixtures in eq. (2.39) gives

$$\begin{aligned} \mathrm{KL}(P : Q) &\le \sum_{j=1}^{M} \lambda_j \,\mathrm{KL}(P_j : q_j) \\ &= \sum_{j=1}^{M} \lambda_j C_d(a_j, b_j), \end{aligned}$$

which proves the bound. Its stated invariances follow term by term from theorem 2.16. □

Corollary 2.22 (Uniform cubical-complex approximation)**.** *Under the assumptions of theorem 2.21, suppose $0 < \mathcal{H}^d(|\mathcal{K}|) < \infty$. On each maximal cell let the tangential law be uniform, i.e. $a_{j,i} = b_{j,i} = 1$ for all i, j, and choose the volume weights*

$$\lambda_j = \frac{\mathrm{vol}_d(C_j)}{\mathcal{H}^d(|\mathcal{K}|)}. \tag{2.41}$$

Let $P = \sum_j \lambda_j P_j$ be the resulting mixture after independent normal Gaussian smoothing on each cell, and let $Q = \sum_j \lambda_j q_j$ be the mixture of the componentwise Gaussian information projections. Then

$$\mathrm{KL}(P : Q) \le \frac{d}{2} \log\left(\frac{\pi e}{6}\right). \tag{2.42}$$

The bound depends only on the intrinsic dimension d and is independent of the ambient dimension, the geometry and embedding of the cubical complex, and the normal Gaussian parameters.

Proof. By proposition 2.6, the volume weights are positive and sum to one. By corollary 2.17, every component divergence equals $\frac{d}{2}\log(\pi e/6)$. Substitution into theorem 2.21 proves eq. (2.42). □

Remark 2.23 (What the mixture bound does and does not assert)**.** *The mixture in theorem 2.21 is obtained by projecting each component separately and retaining the original mixture weights. The theorem proves a constructive upper bound for this particular Gaussian mixture. It does not assert that Q globally minimizes $\mathrm{KL}(P : \widetilde{Q})$ over all Gaussian*

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mixtures $\widetilde{Q}$, nor that the displayed upper bound is the optimal value of such a global mixture optimization problem.

Remark 2.24 (Uniform measure versus its smoothed version)**.** *The volume-weighted identity in* **??** *is an identity of probability measures with respect to d-dimensional Hausdorff measure on the cubical complex. If $d < D$, this unsmoothed measure is singular with respect to D-dimensional Lebesgue measure. The KL guarantee in corollary 2.22 concerns the mixture after adding the componentwise normal Gaussian perturbations, which produces ambient Lebesgue densities; it is not a finite ambient-density KL statement for the original singular uniform measure.*

3. Conclusion

We established a closed-form characterization of the forward Gaussian information projection of Beta–Gaussian distributions supported on affine embedded hypercubes. The optimal Gaussian is uniquely determined by moment matching, and the corresponding minimum KL divergence depends only on the intrinsic Beta parameters, independently of the embedding geometry and ambient dimension. We further extended the result to finite, pure embedded cubical complexes by constructing Gaussian mixtures through componentwise information projections. Joint convexity of the KL divergence yields an explicit approximation bound. In the uniform case, this bound reduces to a linear function of the intrinsic dimension, independently of the geometry, embedding, and ambient dimension.

These results provide a dimension-independent framework for quantifying Gaussian and Gaussian-mixture approximation errors on embedded geometric structures.

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Symbol Legend
⊛: Idea, □: Math, #: Programming, ¶: Writing, ⊥: Critique

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